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What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 2: Two-Oscillator Entropy

Group-B

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1 Normal modes and the reduced state

We first test the entanglement calculation using two coupled harmonic oscillators, whose ground state can be obtained exactly. In units $\hbar = 1$ and with both oscillator masses set to unity,

$$H = \frac{\boldsymbol{p}^T\boldsymbol{p}}{2} + \frac{1}{2}\boldsymbol{x}^T K \boldsymbol{x}, \qquad K = \begin{pmatrix} 1 & c \\ c & 2 \end{pmatrix}. \tag{1}$$

Diagonalising K gives

$$\lambda_\pm = \omega_\pm^2 = \frac{3 \pm \sqrt{1+4c^2}}{2}. \tag{2}$$

A stable ground state requires both eigenvalues to be positive. Since $\operatorname{Tr} K = 3 > 0$, this is equivalent to $\det K = 2 - c^2 > 0$, or $|c| < \sqrt{2}$, satisfied by $c = 0.3, 0.6$. Writing $K = U\,\mathrm{diag}(\omega_+^2, \omega_-^2)U^T$, introduce normal coordinates $\boldsymbol{y} = U^T\boldsymbol{x}$ and momenta $\boldsymbol{\pi} = U^T\boldsymbol{p}$. Then

$$H = \sum_{j=\pm}\left(\frac{\pi_j^2}{2} + \frac{\omega_j^2 y_j^2}{2}\right). \tag{3}$$

Each normal mode has ground-state wavefunction $\psi_0(y) \propto e^{-\omega y^2/2}$, with $\langle y^2\rangle = 1/(2\omega)$ and $\langle \pi^2\rangle = \omega/2$. Transforming back gives

$$X = \langle \boldsymbol{x}\boldsymbol{x}^T\rangle = \tfrac{1}{2}K^{-1/2}, \quad P = \langle \boldsymbol{p}\boldsymbol{p}^T\rangle = \tfrac{1}{2}K^{1/2}, \quad \tfrac{1}{2}\langle\{x_i, p_j\}\rangle = 0. \tag{4}$$

The normal-mode ground state is a product state, but each normal mode involves both original oscillators when $c \neq 0$. Their joint state $\rho_{12} = |\Psi_{12}\rangle\langle\Psi_{12}|$ is pure; tracing out oscillator 2 gives $\rho_1 = \mathrm{Tr}_2\rho_{12}$ with covariance

$$V_1 = \mathrm{diag}(X_{11}, P_{11}), \qquad \nu = \sqrt{\det V_1} = \sqrt{X_{11}P_{11}}. \tag{5}$$

Here $\nu = 1/2$ describes a pure reduced state, while $\nu > 1/2$ indicates mixedness and, because the joint state is pure, entanglement with oscillator 2. Its von Neumann entropy is

$$\boxed{s(\nu) = (\nu + \tfrac{1}{2})\ln(\nu + \tfrac{1}{2}) - (\nu - \tfrac{1}{2})\ln(\nu - \tfrac{1}{2}).} \tag{6}$$

Using $0\ln 0 = 0$, we obtain $s(1/2) = 0$. Entropy is measured in nats.

For an explicit evaluation, set $d = \sqrt{2 - c^2}$ and $t = \sqrt{3 + 2d}$. The matrix square roots are

$$K^{1/2} = \frac{K + dI}{t}, \qquad K^{-1/2} = \frac{1}{dt}\begin{pmatrix} 2+d & -c \\ -c & 1+d \end{pmatrix}, \tag{7}$$

so

$$\nu = \sqrt{\frac{(2+d)(1+d)}{4d(3+2d)}}. \tag{8}$$

Coupling c	ν	Entropy (nats)
0	0.500000000	0
0.3	0.502816542	0.0193599304
0.6	0.512481339	0.0672711138

2 Numerical checks and interpretation

Diagonalisation gives $\nu = 0.5028165419863$ and $s = 0.0193599304236$ at $c = 0.3$. At $c = 0$ the oscillators decouple and the entropy vanishes; increasing c to 0.6 raises it to 0.06727.

As an independent check, use the ground-state Gaussian wavefunction

$$\psi(\boldsymbol{x}) \propto \exp\left(-\tfrac{1}{2}\boldsymbol{x}^T A \boldsymbol{x}\right), \qquad A = K^{1/2}. \tag{9}$$

Tracing out oscillator 2 and evaluating the Gaussian integrals gives the reduced purity,

$$\operatorname{Tr}\rho_1^2 = \sqrt{\frac{\det A}{A_{11}A_{22}}} = \frac{1}{2\nu}. \tag{10}$$

Purity is one for a pure state and smaller than one for a mixed state, independently checking the covariance result.

Under a local canonical rescaling $x_1 \mapsto bx_1$, $p_1 \mapsto p_1/b$, individual variances change, but $X_{11}P_{11} = \nu^2$ and the entropy remain unchanged. The largest numerical change in this product over $b = 0.1, 0.5, 1, 2, 10$ is 5.551×10^{-17}. Thus local squeezing cannot create entanglement between initially decoupled oscillators.

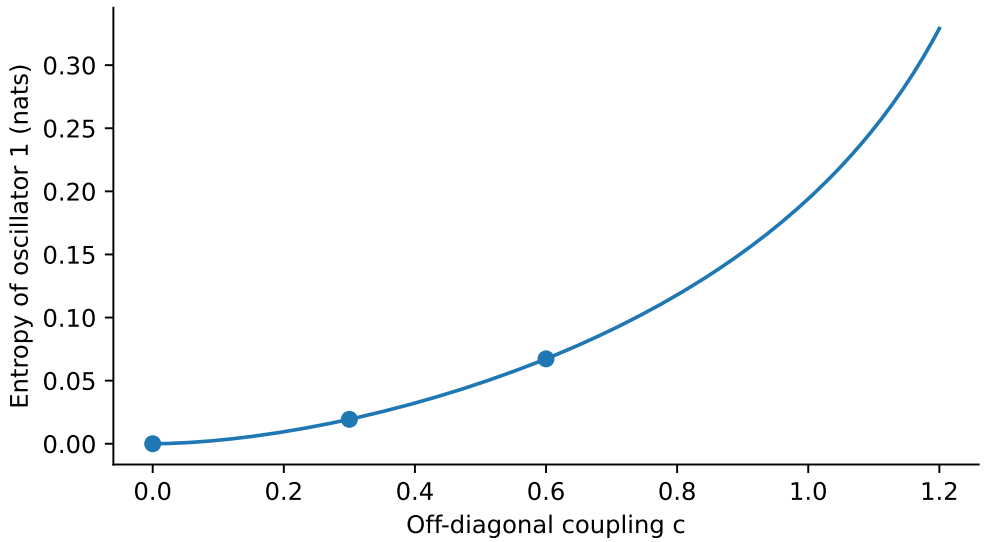


Figure 1: Entanglement entropy of one oscillator versus coupling. The marked points show the decoupled case and the two assigned benchmarks.

For real fields, $q_{-\boldsymbol{k}} = q_{\boldsymbol{k}}^\dagger$. A generic nonzero pair $\pm\boldsymbol{k}$ splits into independent real cosine and sine sectors, each containing one light and one spectator oscillator. Since these sectors are identical and independent in our calculation,

$$S_{\boldsymbol{k},-\boldsymbol{k}} = 2s(\nu_q). \tag{11}$$

We quote the entropy per real sector unless stated otherwise.

References

[1] C. Weedbrook et al., *Gaussian Quantum Information*, Rev. Mod. Phys. 84, 621 (2012), arXiv:1110.3234.

Code and supporting data: step2 on GitHub.

Contributions

Member	Contribution
Priyansh Rawat	Normal-mode derivation and benchmark review.
Samarth Bhatnagar	Reduced-state covariance and rescaling checks.
Mayank Sinha	Numerical diagonalisation, purity cross-check and figure.